Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

To calculate the current drawn by the headlightbulb, we can use the formula: I = P/R, where I is the current, P is the power dissipated, and R is the resistance. Rearranging the formula to solve for R, we get R = P/I. We know the power dissipated is 40 watts, so we can plug that value into the formula: R = 40 / I. We need to find the resistance, so we'll choose the option that gives us a resistance value that, when used in the formula, results in a current value close to 40 watts.